

Database Training with Microsoft SQL Server (DAY – 1)



Authorized & published by
Itlize Global LLC

Agenda: Day -1

- Software
 - Types of Software Applications
- Database
 - Types of database
 - Why to use Database
 - Logical relational data model and referential integrity concept
 - One to one, one to many, many to many
 - Normalization
 - Fundamental building blocks:
 - Table, columns, constraints
 - SQL server various versions
 - Installation of SQL Server Express on your computer/laptop
 - Transactional database versus analytical database

What is a software

software

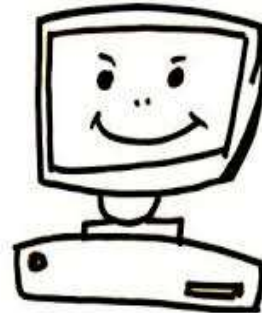
Application
software



System
software



hardware



hardware

- › If hardware is the body, software is the mind
- › In simpler terms, software provides steps by steps instructions for the hardware to operate.

Different type of software applications

- **Web Based Software**

- Software that requires Internet connection to run.
- Examples
 - ❖ Google Maps
 - ❖ Bing Maps
 - ❖ Search Engines
 - ❖ Social Media Applications

- **Windows Based**

- Software that runs native in your machine
- Text editors such as MS-Word
- Anti Virus
- WinZip

- **Apps**

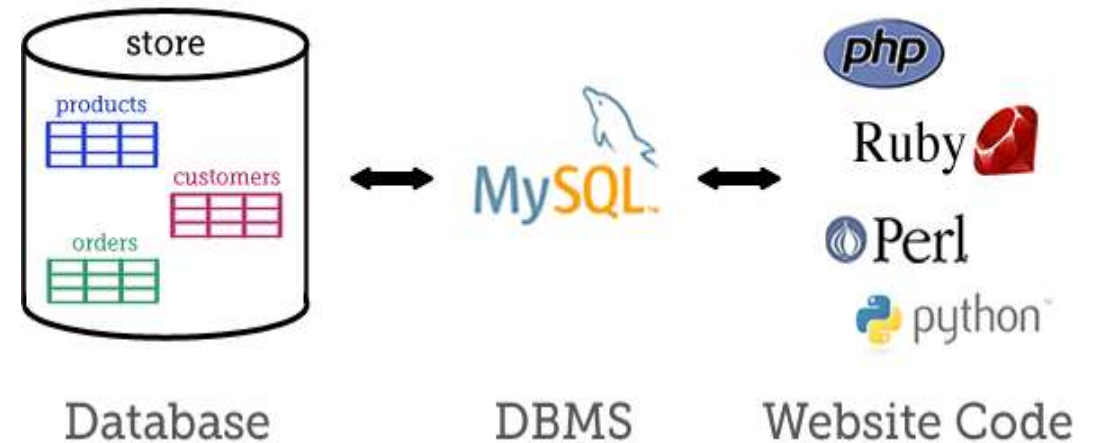
- Designs for Smart Devices such as Tablets or Phones
- IOS/Android & Windows App

- **Business to Business (B2B), Business to Customer(B2C)**

- Web Services/XML/EDI/X12 Standards
- Examples such as payment processing services

What is Database

- A *database* is a collection of information that is organized so that it can easily be accessed, managed, and updated.
 - You can query data in a database in a very efficient way
 - You can relate data from different tables using JOINS.
 - Your data stored using a built-in structure.
 - Databases are ACID(Atomicity, Consistency, Isolation, Durability)



Why to use database

- To eliminate or reduce the data Redundancy
- To remove or reduce the data Inconsistency
- For data Standardization
- Secure all the data in one shell. Performed Role based security
- To make sure that the Integrity of the data is maintained

Different types of databases

- Flat file
 - file containing records that have no structured interrelationship
 - Text files, Comma/Tab Delimited Files
- XML
 - **E**Xtensible **M**arkup **L**anguage
 - Designed to be both human- and machine-readable
- Excel/Spreadsheets
 - Stores data in a tabular form with relationship
- Relational database
 - A database structured to recognize relations among stored items of information.

Logical relationship

- **One to One**

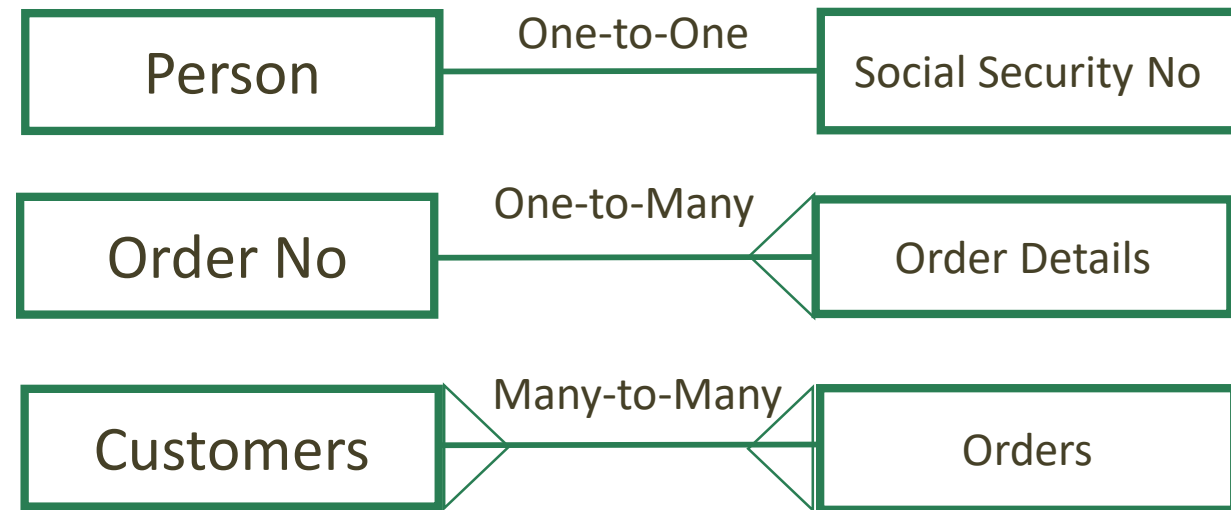
- Both tables can have only one record on either side of the relationship
- Each primary key value relates to only one (or no) record in the related table

- **One-to-many**

- The primary key table contains only one record that relates to none, one, or many records in the related table.

- **Many to Many**

- Each record in both tables can relate to any number of records (or no records) in the other table.



Many to Many Relationship Example

Student ID	Class ID	Student Name
1	3, 5, 9	John
2	1, 4, 5, 9	Debbie



Student ID	Class ID 1	Class ID 2	Class ID 3	Student Name
1	3	5	9	John
2	1	4	5	Debbie



Junction Table

Student ID	Class ID
1	3
1	5
1	9
2	1
2	4
2	5
2	9



Logical relational data model

- In the relational model of a database, all data is represented in terms of tuples, grouped into relations.
- related records are linked together with a "key".

Activity Code	Activity Name
23	Patching
24	Overlay
25	Crack Sealing

Key = 24

Activity Code	Date	Route No.
24	01/12/01	I-95
24	02/08/01	I-66

Date	Activity Code	Route No.
01/12/01	24	I-95
01/15/01	23	I-495
02/08/01	24	I-66

Referential integrity

- Referential integrity means that the foreign key in any referencing table must always refer to a valid row in the referenced table. Referential integrity ensures that the relationship between two tables remains synchronized during updates and deletes

orderNum(Pkey)	customerId	date
1	1	12/27/2015
2	1	12/28/2015
3	2	12/28/2015



orderLineId	orderNum(Fkey)	totalQty
1	1	100
2	1	200
3	2	150
4	3	50

Normalization

Database normalization (or normalization) is the process of organizing the columns (attributes) and tables (relations) of a relational database to minimize data redundancy.

UNF

- All the attributes of the database are simply listed, without any sense of order or grouping.

1NF

- It has an identifying key and it contains no repeating groups of data.

2NF

- It is in 1NF
- All non-key attributes are functionally dependent on the whole key (not part of the key, no partial dependencies)

3NF

- It is in 2NF
- It contains no transitive dependencies

Partial Dependency

- Partial Dependency occurs when a non-prime attribute is functionally dependent on part of a candidate key.
- The 2nd Normal Form (2NF) eliminates the Partial Dependency

StudentID	ProjectNo	StudentName	ProjectName
S01	199	Katie	Geo Location
S02	120	Ollie	Cluster Exploration

Remove Partial Dependency

StudentID	ProjectNo	StudentName
S01	199	Katie
S02	120	Ollie

ProjectNo	ProjectName
199	Geo Location
120	Cluster Exploration

Transitive Dependency

- A transitive dependency in a database is an indirect relationship between values in the same table that causes a **functional dependency**.
- A functional dependency in a database enforces a set of constraints between attributes. This occurs when one attribute in a relation uniquely determines another attribute.
- **A $\rightarrow$ B** which means "B is functionally dependent upon A."
- In this relationship, A determines the value of B, while B depends on A.

Transitive Dependency Example

Author_ID	Author	Book	Author_Nationality
Auth_001	Orson Scott Card	Ender's Game	United States
Auth_001	Orson Scott Card	Children of the Mind	United States
Auth_002	Margaret Atwood	The Handmaid's Tale	Canada

Book → Author

Author → Author_Nationality



This table has a transitive dependency

Book → Author_Nationality

How to remove transitive dependency

Books

Book_ID	Book	Author_ID
Book_001	Ender's Game	Auth_001
Book_001	Children of the Mind	Auth_001
Book_002	The Handmaid's Tale	Auth_002

Book_ID → Book

Authors

Author_ID	Author	Author_Nationality
Auth_001	Orson Scott Card	United States
Auth_002	Margaret Atwood	Canada

Author_ID → Author

Author → Author_Nationality

Author_ID → Author_Nationality

Removing the transitive dependencies(3NF)

Countries

Country_ID	Country
Coun_001	United States
Coun_002	Canada

Authors

Author_ID	Author	Country_ID
Auth_001	Orson Scott Card	Coun_001
Auth_002	Margaret Atwood	Coun_002

Why Transitive Dependencies Are Bad

Author_ID	Author	Book	Author_Nationality
Auth_001	Orson Scott Card	Ender's Game	United States
Auth_001	Orson Scott Card	Children of the Mind	United States
Auth_002	Margaret Atwood	The Handmaid's Tale	Canada

Normalization(Example)

First Normal Form

TABLE: STUDENTS

Name	Age	Pet	Pet Name
Heather	10	Dog	Rex
Heather	10	Cat	Thomas
Rachel	10	Cat	Fluff
Jimmy	11	Dog	Kimba
Lola	10	Cat	Thomas

Second Normal Form

TABLE: STUDENTS

Name	Age
Heather	10
Rachel	10
Jimmy	11
Lola	10

TABLE: PETS

Type	Pet Name	Owner
Dog	Rex	Heather
Cat	Thomas	Heather
Cat	Fluff	Rachel

Normalization(Example)

Third Normal Form

TABLE: STUDENTS

ID	Name	Age
00	Heather	10
01	Rachel	10
02	Jimmy	11
03	Lola	10

TABLE: PETS

ID	Type	Pet Name	Owner ID
00	Dog	Rex	00
01	Cat	Thomas	00
02	Cat	Fluff	01
03	Dog	Kimba	02
04	Cat	Thomas	03

Fourth Normal Form

TABLE: STUDENTS

ID	Name	Age
00	Heather	10
01	Rachel	10
02	Jimmy	11
03	Lola	10

TABLE: PETS

ID	Type	Pet Name
00	Dog	Rex
01	Cat	Thomas
02	Cat	Fluff
03	Dog	Kimba

Table: PETS-STUDENTS

Pet ID	Owner ID
00	00
01	00
02	01
03	02
01	03

Denormalization

- **Denormalization** is a strategy that database managers use to increase the performance of a database infrastructure. It involves adding redundant data to a normalized database to reduce certain types of problems with database queries that combine data from various tables into a single table.
- The main purpose of denormalization is to significantly speed up data retrieval

Fundamental building blocks

- Table
 - A table can store a mass Information in a tabular form
- Fields
 - The information of a table stored against some heads, those are fields or columns. Columns shows vertically in a table.
- Column Names
 - Each fields or column has an individual name. A table cannot contain same name of two different columns
- Record
 - All the columns in a table makes a row. Each row contain all the information of an individual topics.
- Column Value
 - Value of each field makes a row is the column value.
- Key Field (Primary key and Foreign key)
 - Each table should contain a field which can create a link with another one or more table is the key field of a table.

SQL Server Versions

- Enterprise
 - SQL Server Enterprise delivers comprehensive high-end datacenter capabilities for demanding database and business intelligence requirements.
- Business Intelligence
 - SQL Server Business Intelligence empowers organizations to build and deploy secure, scalable and manageable self-service corporate BI solutions.
- Standard
 - SQL Server Standard provides core data management and business intelligence capabilities for non-critical workloads with minimal IT resources.
- Express
 - SQL Server Express is a free edition of SQL Server ideal for developing and powering desktop, web and small server applications.

SQL Server Releases

- SQL Server 2016
- SQL Server 2014 SP1 [12.0.4100.1](#)
- SQL Server 2012 Sp1 [11.0.3000.0](#) Sp2 [11.0.5058.0](#)
- SQL Server 2008 R2
- SQL Server 2008
- And so on 2005,2000,7.0
- **How to: Install SQL Server Express:**

<https://technet.microsoft.com/en-us/library/ms143722%28v=sql.90%29.aspx>

Installing SQL Server Express

- To begin installation, double-click sqlexpr.exe.
- On the End User License Agreement page, read the license agreement, and then select **I accept the licensing terms and conditions** check box. Click **Next**.
- The Welcome to the Microsoft SQL Server Installation Wizard page appears. Click **Next**.
- On the System Configuration Check page, the computer is scanned for potential installation problems. To interrupt the scan, click **Stop**. To proceed with setup after the scan completes, click **Continue**.
- On the Registration Information page, enter information in the **Name** and **Company** text boxes. Click **Next**.
- On the Feature selection page, select the program features to install and click **Next**.
- On the Instance Name page, select a **Default instance** or a **Named instance** for your installation. If you select **Default instance**, an existing default instance will be upgraded. If you select **Named Instance**, specify an instance name or the default named instance of **SQLExpress** is used. Click **Next**.
- The **Service Account** page lets you assign local system account or a domain user account to SQL Server service. This page does not appear if you select the **Hide advanced configuration options** check box

Installing SQL Server Express

- On the **Authentication Mode** page, specify the security mode used to connect to your instance of SQL Server. If you select Windows Authentication, Setup creates an sa account that is disabled by default. Click **Next**.
- On the **Collations Settings** page, change the default **Collation designator and sort order** only if you must match collation settings in another instance of SQL Server or in another computer. Select **SQL collations** to match the sort order settings in earlier versions of SQL Server. Click **Next**. This page does not appear if you select the **Hide advanced configuration options** check box.
- On the **User Instances** page, specify if you want to generate a separate instance of the Database Engine for non-administrator users. By default, user instance functionality is enabled. To turn off user instances, clear the check box. This page does not appear if you select the **Hide advanced configuration options** check box. Click **Next**.
- On the **Error and Usage Report Setting** page, select whether to turn on error reporting and usage reporting for SQL Server and its components. By default, error reporting is turned on. To turn off error reporting, clear the check box. Click **Next**.
- On the **Ready to Install** page, click **Install** to finish installing SQL Server.
- On the **Setup Progress** page, you can monitor installation progress as setup proceeds. Click **Next**.
- On the **Completing the Microsoft SQL Server Installation Wizard** page, you can view the setup summary log by clicking the link provided on this page. To complete the SQL Server Installation Wizard, click **Finish**.

Transactional vs analytical database

- A **transactional database** is a database that allows for **CRUD** actions (Create, Read, Update, Delete) to be taken in a transaction way to make atomic changes.
- An **analytical database** is a database of data which is generally **read-only** and used to store current and historical data for the purpose of mining, generating statistics, projections, and otherwise analyzing it for a certain pattern or criteria.

Transactional vs analytical database

	OLTP System <u>Online Transaction Processing</u> (Operational System)	<u>Analytic Data Warehouse</u>
Source of data	Operational data; OLTPs are the original source of the data.	Consolidation data; OLAP data comes from the various OLTP Databases
Purpose of data	To control and run fundamental business tasks	To help with planning, problem solving, and decision support
What the data	Reveals a snapshot of ongoing business processes	Multi-dimensional views of various kinds of business activities
Inserts and Updates	Short and fast inserts and updates initiated by end users	Periodic long-running batch jobs refresh the data
Queries	Relatively standardized and <u>simple</u> queries Returning relatively few records	Often complex queries involving aggregations
Processing Speed	Typically very fast	Depends on the amount of data involved; batch <u>data refreshes</u> and complex queries may take many hours; query speed can be improved by creating indexes
Space Requirements	Can be relatively small if historical data is archived	Larger due to the existence of aggregation structures and history data; requires more indexes than OLTP
Database Design	Highly normalized with many tables	Typically de-normalized with fewer tables; use of star and/or snowflake schemas
Backup and Recovery	Backup religiously; operational data is critical to run the business, data loss is likely to entail significant monetary loss and legal liability	Instead of regular backups, some environments may consider simply reloading the OLTP data as a recovery method